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Research on PCA Data Dimension Reduction Algorithm Based on Entropy Weight Method

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Abstract—**Traditional principal component analysis (PCA) is a common linear dimension reduction algorithm, but its dimension reduction effect is relatively poor, and the algorithm takes a long time, and it can not meet the prediction target well when applied to the actual scene. Therefore, by introducing the concept of mutual information in information theory and combining with the idea of entropy weight method, an improved PCA algorithm ew-pca is proposed. Firstly, mutual information threshold is set for feature screening, and then the concept of weighted average value is proposed to improve the data centralization process. Finally, the entropy weight is introduced to improve the principal component to optimize the dimensionality reduction process. KNN and SVM algorithm are used to predict and analyze the processed data set. Compared with the traditional PCA algorithm, the improved ew-pca algorithm has better dimension reduction effect and higher prediction accuracy.**

Keywords- data reduction; improved PCA; nutual Information; entropy weight method; feature selection

I. INTRODUCTION

In the field of machine learning, multivariate high-dimensional large data sets not only provide rich information for research and application, but also greatly increase the workload of data processing. Therefore, how to effectively reduce the dimension of data is a research hotspot in the current situation. This paper carries out the exploratory experiment based on the linear dimension reduction method PCA. In recent years, many scholars have carried out a series of related improvements based on traditional PCA[,]. Sun Ping'an et al. Introduced the specific processing process and principle of PCA algorithm in detail[], and verified that PCA may lose some useful information and be easily affected by noise; Zhou Yuhuan and others carried out a research on the effectiveness of feature dimensionality reduction algorithm based on PCA and CCA[], which improved the dimensionality reduction efficiency in subsequent classification, but its applicability was not strong; Sharman et al. Combined PCA with feature ranking[], which proved that the dimension of data set was effectively reduced, but the efficiency was not high; Liu Wenhui and others introduced the idea of mutual information[], and replaced the original PCA covariance matrix by mutual information matrix, the effect of dimension reduction was relatively improved, but the interpretability of features was relatively low. In view of the above problems, this paper proposes an improved PCA dimension reduction method based on entropy weight method, which not only considers the difference of the influence degree of different features on samples, but also improves the accuracy of the principal component model. Experiments show that this method has good dimensionality reduction effect, and the extracted features are more conducive to the prediction task.

II. RESEARCH ON IMPROVED PCA DATA DIMENSION REDUCTION ALGORITHM BASED ON ENTROPY WEIGHT METHOD

A. *EW-PCA process design*

Principal component analysis (PCA) is a linear dimensionality reduction algorithm with the widest application range. This paper combines the idea of entropy method to improve the original PCA algorithm. An improved dimensionality reduction algorithm EW-PCA based on the weight of entropy weight method is proposed. The algorithm first quantifies the data set to obtain the initial matrix.

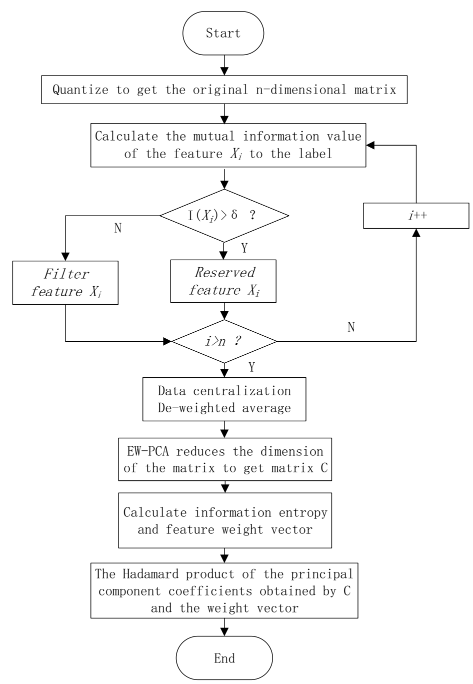

Figure 1. Dimension reduction process of EW-PCA algorithm

The algorithm first quantifies the data set to get the initial matrix. Then calculate the mutual information value of the feature to the label, and use the mutual information threshold to

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filter the matrix. Then use feature weighted average to improve the data centering process. The covariance matrix is obtained through linear transformation, and then the corresponding eigenvalues and eigenvectors are obtained. The number of principal components is recorded as K, and the value of K is determined according to the cumulative contribution rate criterion. Continue the dimensionality reduction process to calculate the information entropy value of each feature. Thus, the weight vector of each feature can be obtained, and the principal component coefficients are weighted twice according to assigned weight. The specific process is shown in Figure 1.

B. Algorithm design

This paper proposes the EW-PCA algorithm, which is improved and extended on the basis of the traditional PCA algorithm. The main steps of the EW-PCA algorithm are:

The first step: feature screening based on mutual information

The main purpose of the PCA dimensionality reduction algorithm based on mutual information is to set the mutual information threshold and filter out some almost useless initial data features, as shown in Figure 2:

Figure 2. Feature selection

Calculate the mutual information values of the remaining features in the processed data set to the tags. The mutual information value range is between [0,1]. Set the mutual information threshold of the data set according to the actual situation, and filter out the features with the mutual information value below the given threshold that have a low contribution rate to the label result, that is, pass the given The threshold realizes the preliminary feature screening process. When the useful and useless features of the data have a clear demarcation point for the mutual information value of the label, it is better to filter based on the importance of the feature to the application analysis.

Step 2: Data centralization processing

Centralize the data according to the weighted average. For each feature of the sample, subtract the weighted average D_n of the feature in the data set from the current value, $X' = x_m^{(n)} - D_n$, $x_m^{(n)}$ represents the nth feature of the mth sample, and the weight distribution basis is each The variance of the characteristic variable.

Among them, the weighting process of each sample is shown in Figure 3:

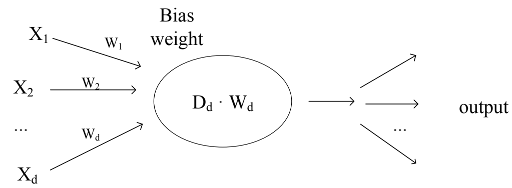

Figure 3. Sample weighting process

For the original matrix transformation processing, the obtained covariance matrix diagonal numerical calculation:

$$\left[\frac{E(x)}{\mathrm{var}(x)}\right]^2 \tag{1}$$

This value can reflect the variation of data in the degree of dispersion on the basis of preserving the correlation between features. The covariance matrix generated can well express all the information of features. Therefore, the weighted average value improvement of traditional PCA is effective and feasible to eliminate the influence of dimension among features.

Step 3: Combine the entropy method to improve the PCA algorithm by weighting the principal components

The improved PCA dimensionality reduction is continued for the matrix generated after sample feature de-weighted average (Data centralization) processing and mutual information feature screening.

(1) Calculate the covariance of the n-dimensional feature data set:

Get the covariance matrix, denoted as Σ:

$$\Sigma = cov(x_1, x_2, \dots, x_n) = \begin{pmatrix} \mathrm{cov}(x_1, x_1) & \cdots & \mathrm{cov}(x_1, x_n) \\ \vdots & \ddots & \vdots \\ \mathrm{cov}(x_n, x_1) & \cdots & \mathrm{cov}(x_n, x_n) \end{pmatrix} \tag{2}$$

(2) Find eigenvalues and eigenvectors:

Solve the characteristic equation $|\lambda E - \Sigma| = 0$, find the characteristic root $\lambda_1 > \lambda_2 > \cdots > \lambda_p > 0$, and the corresponding orthogonalized unit characteristic vector $\alpha_1, \alpha_2, \dots, \alpha_p$, which represent the coefficients on each vector in the new coordinate system[]. The coefficient of the principal component F1 is the unit feature vector α_i.

(3) Determine the number K of PCA principal components according to the CPV criterion of cumulative contribution rate.

Contribution rate:

$$\alpha_i = \frac{\lambda_i}{\sum_{i=1}^{n} \lambda_i} \tag{3}$$

This formula represents the proportion of each feature value in the cumulative sum of feature values, which can be understood as the amount of information that each principal component occupies.

Cumulative contribution rate:

$$a = \frac{\sum_{i=1}^{k} \lambda_i}{\sum_{i=1}^{n} \lambda_i} \tag{4}$$

This formula represents the proportion of the sum of the first k eigenvalues in the overall, which can be understood as the amount of information of the first K principal components in the overall.

The value of α indicates the ability of the first k principal components to synthesize n original features, and reflects the accuracy of the principal component model. The determination of K in this paper selects the value when the cumulative contribution rate is greater than 85, which is considered to be sufficient to reflect the information of the original variable, and realize enough original information to be expressed with fewer components. The expression of the principal component is: $F_i = \alpha_{1i}Y_1 + \alpha_{2i}Y_2 + \cdots + \alpha_{ni}Y_n$. Get the matrix composed of the first k corresponding eigenvectors in descending order.

(4) Weight the principal component coefficients.

It is proposed to weight the principal component coefficients based on the entropy weight method, and perform the matrix Hadamard product calculation on the generated principal component coefficients and the corresponding feature weights obtained by the entropy weight method.

Can be expressed as

$$E^* = E \cdot W \tag{5}$$

$E = [e_1, e_2, \dots, e_k]_{n\times k}$ is the main component coefficient, which is the eigenvector corresponding to the first K eigenvalues, and $W = [w, w_2, \dots, w_n]$ is the obtained weight vector.

III. Experimental Process and Result Analysis

A. The original data set situation

This paper selects a dataset for breast cancer diagnosis of Breast Cancer. The dataset contains 570 data samples, and each sample contains 32 physiological characteristics such as identity ID and diagnosis information; In addition, in order to fully verify the effectiveness of the EW-PCA algorithm, this paper also selects the Sonar and Arcene data sets in the classic UCI library as the input data set for algorithm verification. Sonar has a total of 208 observations, and each example contains 60-dimensional features with a value range of 0 to 1. There are 1-dimensional label columns including R and M categories. Arcene is a data set that identifies cancer based on observational data. This data set is one of the data sets in the NIPS2003 Feature Selection Challenge Competition. It has 700 samples in total and has a data dimension of 10,000.

B. Experimental program

Experiment 1. Dimensionality reduction results

The PCA and EW-PCA methods are applied to the data set to perform dimensionality reduction experiments. Among them, the PCA and EW-PCA methods determine the K value to calculate the contribution rate according to the CPV criterion.

Arcene's dimensionality reduction results are shown in Table I:

TABLE I. Dimensional reduction results of Arcene dataset using PCA and EW-PCA

Algorithm name	contribution rate				
	0.80	0.85	0.90	0.95	0.99
PCA	8	14	34	157	485
EW-PCA	2	3	6	63	297

It can be observed from Table I that when the contribution rate is 95%, the K value of the PCA method is 157, and the K value of EW-PCA is 63. It can be seen that in the case of uniform information distribution, compared with traditional PCA Method, through EW-PCA dimensionality reduction, fewer principal components can be used to retain as much useful information as possible in the original data set. In the case of the same contribution rate, it can be seen from the number of principal components that the dimensionality reduction effect of EW-PCA is significantly better than the traditional PCA method.

For the convenience of visual display, draw a line graph of the dimensionality reduction result as shown in Figure 4:

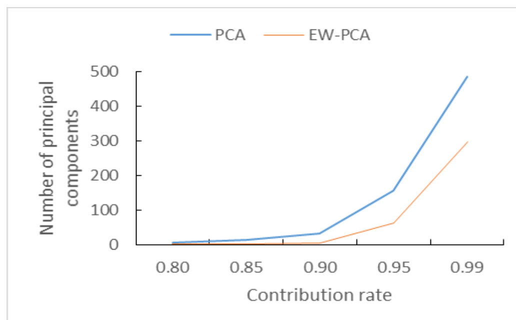

Figure 4. PCA and EW-PCA dimensionality reduction results of Arcene

It can be seen from Figure 4 that the number of principal components of the dimensionality reduction results of the traditional PCA method and the EW-PCA method both increase with the increase of the contribution rate.

In order to compare the difference of the principal component directions obtained by the traditional PCA method and the EW-PCA method of dimensionality reduction, the two methods were applied on the Sonar data set to conduct a dimensionality reduction comparative experiment. Table II intercepts the principal component coefficients and cumulative contribution rate of the first seven principal components. Observing the two methods when the principal components are the same, the principal component contribution rate of EW-PCA is greater than the result obtained by PCA. The experiment shows that the method of weighting and improving the principal components by calculating the information entropy combined with the entropy weight method is reasonable, and the cumulative contribution rate obtained by the EW-PCA method is greater.

TABLE II. PRINCIPAL COMPONENT COEFFICIENTS AND CUMULATIVE CONTRIBUTION RATES OBTAINED BY PCA AND EW-PCA

Algorithm	Principal	Principal component coefficient	Cumulative
PCA	1	[-6.663e-04, …… ,4.202e-04]	31.97
	2	[1.463e-02, …… ,2.816e-03]	52.35
	3	[6.034e-03, …… ,5.194e-04]	60.91
	4	[1.656e-02, …… ,7.779e-04]	67.37
EW-PCA	1	[1.025e-04, …… ,-7.14e-06]	35.58
	2	[-1.299e-03, …… ,-5.727e-05]	57.72
	3	[5.867e-04, ……,1.211e-05]	67.45
	4	[3.254e-04, …… ,2.484e-06]	73.15

In order to compare the performance of the traditional PCA method and the EW-PCA method on different data sets, the running time and K value under the same contribution rate are listed. Obviously, on the Breast Cancer and Sonar data sets with relatively low dimensions, the two The method has little effect on the running time, but on the Arcene dataset with higher dimensionality, the time cost of EW-PCA will increase slightly. On different data sets, compared with PCA, the EW-PCA method may sacrifice a small part of time overhead, but the dimensionality reduction effect is better.

TABLE III. PERFORMANCE COMPARISON OF PCA AND EW-PCA RESULTS

Data Set	Algorithm Name	Contribution Rate	Running time（s）	K value
Breast Cancer	*PCA*	0.85	0.078	4
	EW-PCA	0.85	0.072	4
Sonar	*PCA*	0.85	0.091	9
	EW-PCA	0.85	0.099	7
Arcene	*PCA*	0.85	118.53	14
	EW-PCA	0.85	126.76	3

Experiment 2. Comparing prediction accuracy

In order to verify the impact of EW-PCA on the accuracy of prediction, this paper uses KNN and SVM algorithms to test the data sets processed by the two dimensionality reduction algorithms, and apply the two to the breast cancer data set for prediction, PCA and EW-PCA Take the same contribution rate. In order to verify the impact of data dimensionality reduction on the prediction accuracy, and at the same time compare the accuracy of direct prediction with the original data, the results are shown in Table IV:

TABLE IV. COMPARISON OF PREDICTION ACCURACY BETWEEN PCA AND EW-PCA

Algorithm Name	Contribution Rate	Before dimensionality reduction		After dimensionality reduction	
		KNN	*SVM*	*KNN*	*SVM*
PCA	0.95	0.9665	0.9728	0.9603	0.9457
EW-PCA	0.95	0.9665	0.9728	0.9682	0.9770

It can be seen that when KNN is selected for prediction, the accuracy of prediction after dimensionality reduction by the PCA method due to the loss of information is lower than that of directly prediction to the original data, while the improved EW-PCA method is more accurate after dimensionality reduction. Comparing the prediction accuracy of PCA and EW-PCA before and after dimensionality reduction, whether KNN or SVM, the prediction and classification accuracy of PCA reduced breast cancer data set is lower than that before dimension reduction, but after dimension reduction by ew-pca algorithm, the prediction accuracy is slightly higher than that of direct prediction, and the effect is better. Therefore, the improved ew-pca algorithm proposed in this paper has higher prediction accuracy and better prediction task partition effect.

IV. CONCLUSION

In the process of data processing, the traditional linear dimensionality reduction PCA relies solely on the analysis of objective data, the dimensionality reduction results are not good enough, the time-consuming is too long, the information loss is high, and the prediction accuracy of the data after dimensionality reduction is not high. To deal with these problems, this paper proposes an improved PCA dimensionality reduction algorithm EW-PCA based on the weight of the entropy method. The PCA principal component coefficients are weighted and improved by calculating the information entropy value, and the EW-PCA and PCA algorithms are applied to the Breast Cancer dataset and the benchmark dataset Sonar and Arcene. Predictive analysis experiments are carried out using KNN and SVM algorithms. Experiments show that the EW-PCA method is more cautious in finding the principal components, and the feature description is clearer. Compared with traditional PCA, it has better performance and results, and the data set after dimensionality reduction has higher prediction accuracy.

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